ANSHIKA BAJPAI

anshikabajpai23@gmail.com | github.com/anshikabajpai23 | linkedin.com/in/anshikabajpai23/ | bajpaianshika.netlify.app/ | +19302043030

SUMMARY

Machine Learning Engineer and software engineer with 4+ years of experience designing and deploying LLM-powered systems, RAG pipelines, and agentic workflows in production. Strong foundation in MLOps, distributed training, and model monitoring, with research experience in generative models. Proven ability to reduce inference latency, improve retrieval precision, and ship ML systems supporting enterprise-scale users.

EDUCATION

Indiana University Bloomington, Master of Science in Data Science, USA **(GPA: 3.94)** *May 2026*

B. Tech in Computer Science and Engineering, Jaypee Institute of Information Technology, India *May 2021*

WORK EXPERIENCE

Machine Learning research Assistant, Indiana University (AIMed Lab) *Jan 2026 – May 2026*

- Engineered **DESS→PD medical image translation pipeline using RegGAN** on **NVIDIA A100 GPU**, improving distribution alignment by **37% (FID 260→164)** while **preserving anatomy** with **0.055px deformation** and **0% topology violations**, **outperforming CycleGAN baseline** with **40× lower deformation**
- Developed end-to-end **computer vision preprocessing pipeline** for cross-modality MRI alignment across 11,000+ image slices on **HPC cluster**; evaluated using **FID, SSIM, and Jacobian determinant metrics**; extending to **diffusion models (DDPM)** to generate synthetic healthy knee MRI images, enabling before-and-after injury comparison for personalized treatment planning.

Machine Learning Engineer Intern, Palo Alto Networks, California *May 2025 - Aug 2025*

- Built **end-to-end RAG pipeline** on **GCP** (Vertex AI Embeddings + Enterprise Search) for document Q&A serving **10,000+ employees**; implemented **Top-K semantic retrieval** with **query-time caching (Redis)** , cutting latency **75%** (4.8s to 1.2s) and redundant API calls by **41%**
- Architected **async document ingestion pipeline** via **GCP Pub/Sub** for real-time PDF/DOCX/HTML processing; applied **BigQuery** content hashing and metadata deduplication to cut duplicate ingestion by **~35%** across **500+ internal documentation sources**
- Engineered **multi-turn conversational agent** with **context compression** and **sliding-window memory**, enabling **20+ turn** coherent sessions; improved answer relevance by **31%** via **chain-of-thought** and **few-shot prompt templates**
- Designed and ran **A/B retrieval evals** across chunking strategies, **semantic chunking** boosted **Precision@3 by 18%** on cybersecurity queries, quantifying the impact of prompt and retrieval changes

Senior Software Engineer, Optum (UnitedHealth Group) *Jun 2021 – Jun 2024*

- Refactored legacy system reducing processing time from **500+ minutes to under 3 minutes**; optimized error report generation cutting errors from **25,000 to 20** via automated testing and CI/CD integration, **improving latency and reducing production costs**
- Designed and deployed a distributed **microservices-based notification system** for compliance notices, processing millions of notifications and **saving $500K annually** as part of a company-wide Paperless Initiative
- Drove engineering excellence by introducing design patterns, achieving 90%+ unit test coverage, and establishing code review guidelines; mentored junior engineers across teams, fostering a culture of quality and continuous improvement.

Machine Learning Scientist Intern, Taiyo LLC *Dec 2019 – Apr 2020*

- Implemented an end-to-end ***ML pipeline for stock market prediction***, including ***time series forecasting, hyperparameter tuning across 10+ models***, and improving accuracy by **15%** on 50,000+ records.

SKILLS

Programming Language: Python, Typescript, React, R, SQL, C/C++, PostgreSQL, BigQuery, Linux, Unix, ETL, Codex, Claude, OpenAI API

Frameworks/Tools: Jenkins, Git, GCP, DevOps, Docker, Kubernetes,PySpark, LangChain, RAG, Azure, AWS, MLflow, CrewAI, Airflow, Agile

AI/ML: PyTorch, Natural Language Processing (NLP), Neural Networks (including CNN, RNN), Deep Learning, Generative AI(Gen AI), Recommendation System, MLOps, Anomaly Detection, LLMs, Graph NN, TensorRT-LLM, Vertex AI, Prompt Engineering, n8n

Libraries/Software: Numpy, Matplotlib, Pandas, NLTK, scikit-learn, OpenCV, Grafana, Snowflake, Streamlit, FastAPI, Keras, TensorFlow

PUBLICATIONS

- **Bajpai A.,** D. Cavar, et al, Improving LLM Reasoning Through Ontology-driven Knowledge Graph and FAISS s: A Comparative Study of Generating Ontologies for Medical RAGs, Midwest Speech and Language Days 2025, Indiana University Bloomington, April 2025.
- **Bajpai A.**, Garg M., Hindi Sentiment Analysis on Tweets, 2nd International Conference on Artificial Intelligence: Theory & Applications.

PROJECTS

Incident Severity Auto-Triage with RLHF | LLaMA 3.2 1B, QLoRA, GRPO, TRL, PEFT, Hugging Face Transformers, Streamlit, Python, GPU A100

- Fine-tuned LLaMA 3.2 1B with QLoRA on 124K bug reports; diagnosed and fixed token ID mapping bug in weighted cross-entropy loss and gradient dilution bug, lifting **P0 F1 from 0.05 → 0.41** and **team routing accuracy from 0.51 → 0.82**
- Improved **severity macro-F1 from 0.41 → 0.57** across 5 imbalanced classes via mild oversampling and per-class token weighting; reduced **parse failure rate 17.8% → 12%** by fixing train/inference prompt mismatch
- Implementing **RLHF pipeline with FSDP: SFT → Reward Model → GRPO** for incident severity auto-triage on A100 GPUs

FAQ Agent Evaluation Pipeline | Agentic AI, LangGraph, LangChain, LangSmith, OpenEvals, Qwen, Pytest, Python, Claude API, FAISS

- Developed **agentic RAG system** for insurance claims Q&A with **LangGraph, ReAct agents, multi-agent orchestration**, and **hybrid BM25 + vector retrieval**, routing queries across auto, home, and billing agents; implemented **MCP server** to expose retrieval tools as standardized agent interfaces, achieving 83% correctness / 84% faithfulness on LangSmith-evaluated Q&A pairs.
- Built **LLM evaluation framework** with **LLM-as-judge, RAGAS-style metrics, and LangSmith** tracing to evaluate correctness, faithfulness, and retrieval relevance across 60 **adversarial and multi-hop test** cases; surfaced judge inconsistency and score inflation in 7B models, informing retrieval and prompt iteration cycles